

Statistical Optimization of Wire EDM Parameters for Surface Roughness in P-20 Tool Steel Using Orthogonal Array Design, ANOVA, and Regression Modeling

Siddiqui Mohd Abdul Mukhtadir Siddiqui Abdul Rub¹, Brijbhushan Shukla²

1Mtech Student, Shreeyash college of Engineering & Technology, Chhatrapati Sambhajnagar (Aurangabad), Maharashtra, 431010.

2Professor, Shreeyash college of Engineering & Technology, Chhatrapati Sambhajnagar (Aurangabad), Maharashtra, 431010.

Corresponding Author: sabdulmuqtadir007@gmail.com

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Abstract

This study focuses on the optimization of wire cut electrical discharge machining (WEDM) parameters for improving surface roughness (SR) in P-20 tool steel, a widely used material in mold and die applications. The machining process was investigated with three critical input variables pulse-on time, pulse-off time, and current selected based on their strong influence on the electro-thermal erosion mechanism. Experiments were conducted using an L9 orthogonal array designed through the Taguchi method, enabling systematic evaluation with a reduced number of experimental runs. The experimental results were statistically analyzed through analysis of variance (ANOVA) to quantify the contribution of each parameter toward SR. The results demonstrated that pulse-on time was the most significant factor (36.15%), followed by pulse-off time (32.14%) and current (26.34%). A regression model was further developed to predict surface roughness as a function of process parameters. The model exhibited good predictive capability, with less than 10% deviation between predicted and experimental values. A confirmation experiment was carried out at the optimal parameter settings (pulse-on time of 25 μ s, pulse-off time of 12 μ s, and current of 12 A), and the results showed close agreement with the predicted outcomes, with only 4.19% error. The study confirms that the Taguchi approach, combined with ANOVA and regression modeling, is highly effective in optimizing WEDM parameters for achieving superior surface quality. The findings provide useful insights for mold and die industries, where enhanced surface finish directly improves product performance, wear resistance, and service life.

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1. Introduction

Electrical Discharge Machining (EDM) is a significant non-conventional machining process that has taken the place of the conventional machining of hard to machine materials. As the aerospace, automotive, tool, and die components keep gaining demand, tribal metals like titanium, Inconel, ceramics, carbides, and special tool steels are popular because of their excellent mechanical and thermal capabilities. Nevertheless, traditional machining processes may fail or prove to be costly with regard to these materials. The solution is the Wire Cut Electrical Discharge Machining (WEDM) where a continuously moving wire electrode is used to erode the workpiece by controlled

spark discharges in a dielectric fluid. Such a process enables machining with high accuracy of intricate shapes and hard materials. In the contemporary production process, the attainment of a high surface finish has taken on the same significance as the enhancement of productivity. The surface roughness directly influences the performance, wear resistance and service life of components particularly in moulds and dies where high surface quality is required. Various researchers have made lots of contributions to the study of the performance of WEDM on P-20 tool steel. Ugrasen et al. (2014) estimated the machining performances of P-20 material at Group Method Data Handling technique. Priyadarshini et al. (2018, 2022) also examined the WEDM of P-20 tool steel, and used a further method of analyzing surface and tribological characteristics, grey relational optimization. Jain et al. (2017) compared the wear and acoustic emission parameters of the electrode in machining of P-20 in MRA and GMDH methods. Yadav et al. (2014) showed the use of Taguchi approach in the manufacturing of punching dies with the help of P-20 material. Mahapatra et al. (2021) built on the study by using multi-objective optimization in AISI P-20 tool steel at EDM. Together, all these studies demonstrate the significance of optimizing the parameters in order to realize improved machining quality and performance.

Hence, the present study focuses on investigating the influence of WEDM process parameters—pulse on time, pulse off time, and current on surface roughness (SR) of P-20 tool steel. Optimization using the Taguchi method, combined with statistical validation through ANOVA and regression modeling, aims to establish reliable parameter settings for achieving minimal surface roughness.

More recently, researchers have explored advanced approaches to enhance WEDM outcomes. Vijayakumar and Chandradass (2024) employed machine learning techniques to optimize WEDM parameters for gear machining, highlighting the role of predictive algorithms in improving machining accuracy. Somavanshi and Sathe (2023, 2024) investigated electrochemical and nanomaterial-based machining aspects, providing insights into hybrid strategies for better process control. Kumar and Das (2025) studied process parameters in ultrasonic machining, emphasizing the role of statistical design in precision manufacturing. Nath and Xavier (2025) contributed to material characterization studies relevant for machining applications, while Shrivastava and Biswas (2025) focused on catalytic process improvements, reflecting the broader integration of material science with machining technology. These recent contributions underline the growing emphasis on hybrid optimization, machine learning integration, and advanced material considerations in non-traditional machining research.

2 Experimentation

2.1 Methodology of Experiment

There are some optimization methods that can be utilized in the development and enhancement of processes. The typical ones are Build-Test-Fix (BTF), Design of Experiments (DOE), and One Variable at a Time (OVAT). The BTF method is crude and unstructured, relying on corrections made on a trial-by-trial basis. DOE is very effective because it enables the simultaneous rotation of numerous parameters, although it necessitates numerous experimental manifolds and it is too closely connected with the statistical interpretation. OVAT, by contrast, explores the impact of a

single parameter at a time, maintaining all the rest constant, which is clear in the impact of a single parameter at the cost of efficiency.

In the current research, pulse-on time (T-on), pulse-off time (T-off), and current were identified as the most significant parameters affecting the performance of WEDM. The Taguchi method was adopted because it is effective in optimizing processes involving a few experiments in a multifactorial procedure. Previous OVAT experiments were carried out to determine the effective parameter ranges. One of the designs (L9 orthogonal array or OA) was chosen that would allow three parameters to be used, with three levels per parameter, to represent a minimum number of experimental runs while still maintaining statistical reliability. Then the analysis of variance (ANOVA) was used to measure the contribution of each parameter. All calculations and statistical procedures were carried out using Minitab 19 software.

2.2 Experimental Machine Selection

Table 1. WEDM Machine Specification.

Model Name	Job Master ID-Zire
Table Size	635 x 475mm
Work-tank Internal Dimensions	800 x 650
Main Table Travel X, Y	400 x 300
Z-Axis Travel	250
Maximum Taper Angle	$\pm 30 / 50$ Deg/mm
Maximum Work-piece weight	300kg
Wire Diameter	0.15 ~ 0.3mm



Figure 1. Set up a wire-cut electro discharge Machine.

2.3 Selection of material

P20 Mold Steel is a versatile, low-alloy tool steel that is characterized by good toughness at moderate strength levels. The steel is commonly used for plastic injection mold cavities and tooling and for die casting dies for zinc. Size available in round, flat and square. Literature studies indicate that research can be conducted to evaluate the effect of process

parameters, such as pulse time, Pulse of time, and Current of wire cut electro discharge machining, on surface roughness and material removal rate. Chemical composition of P-20 Steel is shown in Table 2

Table 2. Chemical Composition of P-20 Steel.

Composition	C	Si	Mn	Cr	P
Percentage	0.28-0.40%	0.20-0.80%	0.60-1.00	1.40-2.00%	0.03



Figure 2. P-20 Steel.

2.4 Levels of Input Parameters

Three levels for each parameter have been selected for optimization. Selecting more than 3 levels would have needed more experiments to be conducted. Selecting less than 3 levels is not justified for the investigation of the effect of parameters for 1st time. Table 3 shows three levels of input parameters selected for optimization study.

Table 3. Levels of Input Parameters.

Sr. No	Level 1	Level 2	Level 3
Pulse on time (μ s)	25	30	35
Pulse off time (μ s)	8	10	12
Current (A)	12	15	18

3. Results and Discussion

To get a complete understanding of the effects of input parameters, pulse on time, pulse off time, and current on output surface roughness and material removal rate, you usually assess the signal-to-noise ratio or the main effects plot for means. For this purpose, Minitab 19 statistical software has been used. Modeling of surface roughness and material removal rate has been done. ANOVA has been conducted to find out the effect of each parameter on the surface roughness, and a linear regression model has been established to predict the values of surface roughness.

3.1 Experimental Result

Table 3 presents the L9 orthogonal array with measurements of surface roughness for runs 1 to 9. It also shows S/N ratio for all nine experiments.

Table 3. L9 orthogonal array with response characteristic.

Experiments	Inputs Factors			Output Responses	
Trial No.	Pulse on time (μ s)	Pulse off time (μ s)	Current (A)	SR(Ra) (μ m)	S/N Ratio
1	25	8	12	2.89	-9.2180
2	25	10	15	2.65	-8.4649
3	25	12	18	3.01	-9.5713
4	30	8	15	3.86	-11.7317
5	30	10	18	3.12	-9.8831
6	30	12	12	2.68	-8.5627
7	35	8	18	3.68	-11.3170
8	35	10	12	3.39	-10.6040
9	35	12	15	3.20	-10.1030

The S/N ratio values are calculated using Minitab 19 software. It can be seen that the variation in the S/N ratio is minimum for all experiments.

3.2 Main Effects of SR

From the main effects plot for S/N ratio, the parametric effect on the response characteristic, i.e., SR, can be understood. Pulse on time 25 μ sec at level 1, Pulse off time 12 μ sec at level 3, Current 12 A at level 1 gives the highest signal to noise ratio values. Figure 5 shows the main effects plot from S/N ratios. The levels at which the highest S/N ratio is obtained from the S/N ratio plot are taken as optimum levels for setting machine parameters.

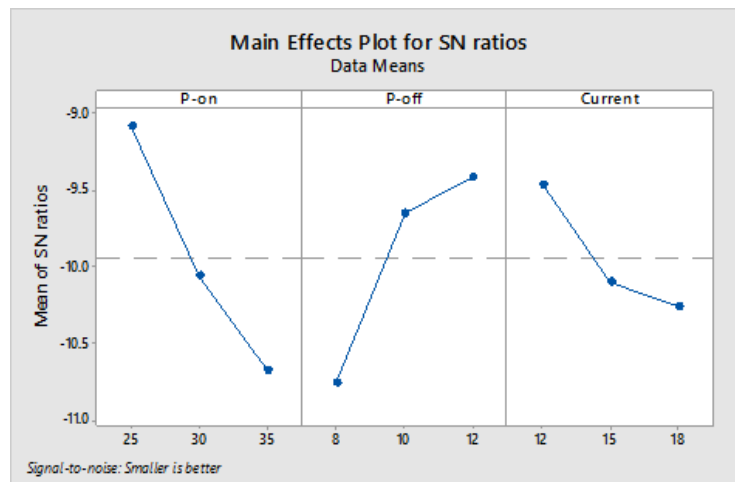


Figure 3. Main Effects Plot for S/N Ratio.

3.3 ANOVA Result

In ANOVA, the ratio between the variance of the cutting parameter and the error variance is called Fisher's ratio (F). It is used to determine whether the parameter has a significant effect on the quality characteristic by comparing the F test value of the parameter

with the standard F table value at the P significance level. If the F-test value is greater than the P-test, the cutting parameter is considered significant. The relevance of the models is tested by analysis of variance (ANOVA). It is a statistical tool for testing the null hypothesis for planned experiments, in which several different variables are studied simultaneously. ANOVA is used to quickly analyze the variances in the experiment using the Fisher test (F test). ANOVA table shows the result of the ANOVA analysis. ANOVA analysis makes it possible to observe that the value of P is less than 0.05 in the three parametric sources.

Table 4. ANOVA Result.

Source	DF	Adj SS	Adj MS	F-Value	P-Value	% Contribution
T-on	2	0.5070	0.2535	6.75	0.017	36.15
T-off	2	0.4508	0.2254	6.01	0.020	32.14
Current	2	0.3694	0.1847	4.91	0.047	26.34
Error	2	0.0751	0.0375			
Total	8	1.4023				

It shows that the pulse on time (36.15%), the pulse off time (32.14%), and the Current (26.34%) have a significant influence on surface roughness. The contribution of P-on (36.15%) is the highest among all three parameters; hence, it is the most dominant parameter, while pulse off time is the least affected parameter.

3.4 Development of a Regression Model for Surface Roughness

A regression model has been developed using Minitab software. Substituting the experimental values of the parameters in the regression equation, values for surface roughness have been predicted for all levels of study parameters. The graphical representation also shows that the predicted and experimental values of surface roughness correlate with each other.

Regression Equation

$$SR = 1.66 + 0.0737 \text{ P-on} - 0.128 \text{ P-off} + 0.0456 \text{ Current}$$

Table 5. Experimental and Predicted Values of SR.

Sr. No.	Experimental value	Predicted value	Error %
1	2.89	3.02	4.30
2	2.65	2.90	8.62
3	3.01	2.78	8.27
4	3.86	3.53	9.31
5	3.12	3.41	8.50
6	2.68	2.88	6.94
7	3.68	4.03	8.68
8	3.39	3.50	3.14
9	3.20	3.38	5.38

Table 5 presents a comparison between experimentally measured and predicted surface roughness values obtained using the developed mathematical equation.

The difference between surface roughness values calculated using the regression equation and experimental values for each experience was less than 10%. Hence, we can conclude that the developed regression equation is valid. Figure 3 shows the graphical representation of experimental and the values calculated using the regression equation.

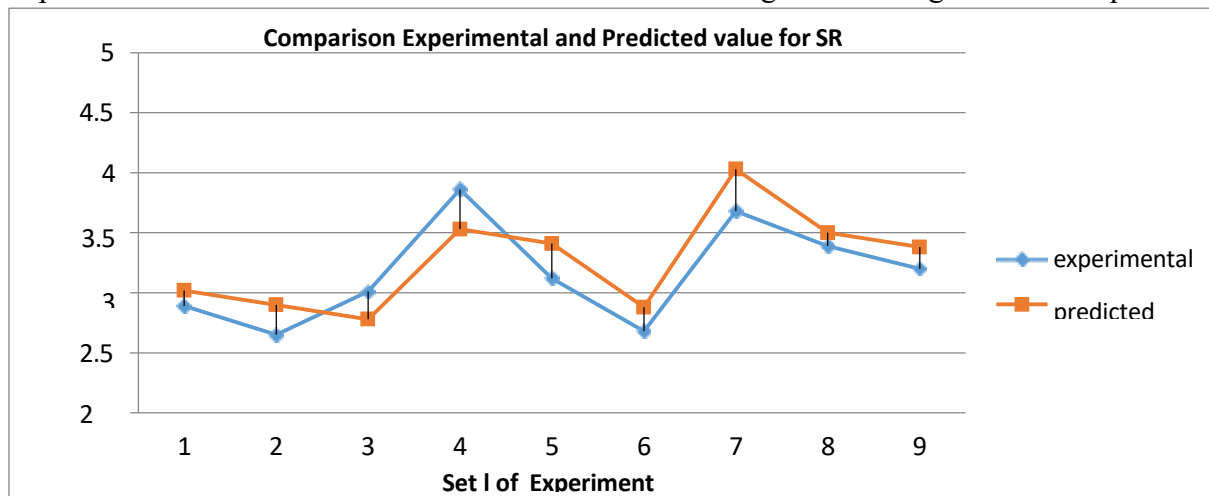


Figure 4. Comparison between Experimental and Predicted values of SR.

3.5 Confirmation Experiment Result

Table 6 shows the difference between the surface roughness value of the confirmation experiment and the value predicted by the regression model developed.

Table 6. Confirmation Experiment Result.

Parameter	Model value	Experimental value	Error %
SR	2.51	2.62	4.19

A confirmation experiment is conducted by keeping parameters at optimum levels suggested by the Taguchi method, and the surface value obtained has been compared with the value predicted by the regression model, keeping the parameters at the same levels. It can be seen that the difference between the experimental result and the predicted result is 4.19%. This indicates that the experimental value correlates with the estimated value.

4. Conclusions

Based on the experimental investigations and statistical analysis carried out in this study, the following conclusions are drawn:

1. Surface roughness (SR) in WEDM is strongly influenced by pulse on time, pulse off time, and current.

2. The optimal parameter combination for achieving minimum SR was found to be: pulse on time of 25 μ s, pulse off time of 12 μ s, and current of 12 A.
3. ANOVA results confirmed that pulse on time contributed the most (36.15%) towards SR, followed by pulse off time (32.14%) and current (26.34%).
4. The developed regression model reliably predicts SR with less than 10% error, providing a useful tool for process planning and control.
5. The confirmation experiment validated the optimization results, showing only 4.19% deviation between experimental and predicted SR values.
6. This research demonstrates that the Taguchi method is an effective approach for optimizing WEDM parameters to enhance surface quality in P-20 tool steel machining.

5. Conflict of interest

None

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